

Expressions: Like Terms & Distributive

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Like terms have the same variable and exponent, so they can be combined: $3x + 5 + 2x = 5x + 5$. The distributive property says $a(b + c) = ab + ac$, so $3(x + 4) = 3x + 12$.

Round 1 — Retrieve It

1. What two things must terms share to be "like terms"?

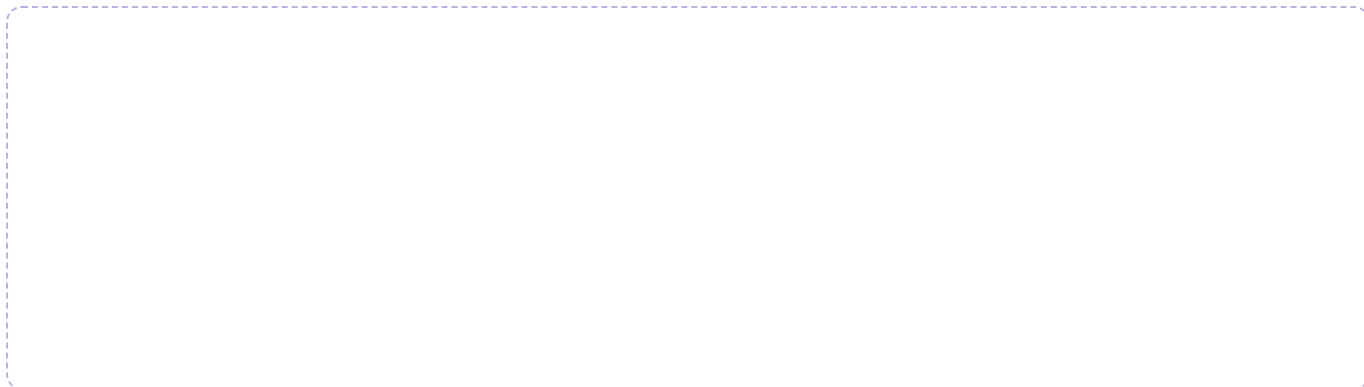
2. Combine the like terms: $3x + 5 + 2x$.

3. Write the distributive property using a , b , and c .

4. Use the distributive property to expand $3(x + 4)$.

Round 2 — Sketch It

Draw the expression $3(x + 4)$ as three groups, each holding one x and four 1 s. Circle the total number of x tiles and the total number of 1 tiles to show why it equals $3x + 12$.



Expressions: Like Terms & Distributive — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Simplify: $7a + 3 + 2a + 5$.

2. Expand: $5(y + 2)$.

3. Expand, then combine like terms: $2(x + 3) + 4x$.

Round 4 — Reason from Evidence

A student wrote $3x + 2 + 4x = 9x$. Use like terms to find the mistake, give the correct simplified expression, and explain why 2 cannot be combined with the x terms.

Expressions: Like Terms & Distributive — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how you know which terms in an expression can be combined. Use one example that is not printed on the cue card.

C**Connect**

Connect the distributive property to finding the area of a rectangle that is split into two parts. How does $a(b + c) = ab + ac$ show up?

E**Extend**

Create your own expression that needs both the distributive property and combining like terms. Show the work and give the simplified result.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Expressions: Like Terms & Distributive

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. The same variable and the same exponent.
2. $3x + 2x + 5 = 5x + 5$.
3. $a(b + c) = ab + ac$.
4. $3(x + 4) = 3x + 12$.

Sketch It — look for

Three equal groups, each with one x and four 1s: 3 x tiles and 12 unit tiles in all, matching $3x + 12$.

Apply It

1. $7a + 2a + 3 + 5 = 9a + 8$.
2. $5(y + 2) = 5y + 10$.
3. $2(x + 3) + 4x = 2x + 6 + 4x = 6x + 6$.

Reason from Evidence — look for

$3x + 4x = 7x$, and the 2 is a constant, so it cannot join the x terms. Correct answer: $7x + 2$. The 2 has no variable, so it is not a like term with $3x$ or $4x$.

ACE — Extend look-for

Many answers possible. Check that the expression uses distribution first, then combines like terms correctly.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.